

Activity 7 Pre-Lab — Grignard: Synthesis of Triphenylmethanol

[Your name] & [partner]
[MM/DD/YY]

Organic Lab · Grignard reagent + addition to a ketone

■ **Black = write this in your notebook** ■ **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Make the **Grignard reagent phenylmagnesium bromide** from bromobenzene + Mg, then use it as a **carbon nucleophile** that adds to the carbonyl of **benzophenone**. Acidic work-up gives **triphenylmethanol**, purified by recrystallization.

👉 **DRAW** — the two steps

1) bromobenzene + Mg → phenylmagnesium bromide (PhMgBr) **dry Et₂O, I₂ to initiate**

2) PhMgBr + benzophenone (Ph₂C=O) → [Ph₃C-O⁻ MgBr⁺] → **triphenylmethanol** (Ph₃C-OH) **then H₃O⁺ (dil. H₂SO₄) work-up**

CONTEXT ▶ ALL glassware must be bone-dry — water quenches (destroys) the Grignard. Don't wash glassware the day of lab.

Chemical Reagents & Safety Table

CONTEXT ▶ Required: magnesium, iodine, diethyl ether, bromobenzene, benzophenone, sulfuric acid. Typical SDS values — cross-check the SDS tile.

Chemical	Role	MW	Amount used	Physical properties	Hazards (GHS)
Magnesium (turnings)	Makes the Grignard	24.31	0.5 g (~0.021 mol)	Silvery metal solid	H228 flammable solid · H261 contact w/ water → flammable gas
Iodine	Initiator (activates Mg)	253.81	1 small crystal	Purple-black solid; sublimes	H312+H332 harmful (skin/inhaled) · H315 · H319 · H335 · H372 (thyroid) · H400 aquatic
Diethyl ether (anhydrous)	Dry solvent	74.12	12 + 5 + 30 mL	Colorless liquid; bp 34.6 °C; d 0.71 g/mL; flash pt -45 °C	H224 extremely flammable · H302 · H336 · forms explosive peroxides
Bromobenzene	Grignard precursor (limiting)	157.01	2.0 mL (~3.0 g, ~0.019 mol)	Colorless liquid; bp 156 °C; d 1.50 g/mL; flash pt 51 °C	H226 flammable · H315 skin irritation · H411 aquatic tox
Benzophenone	Electrophile (ketone)	182.22	3.2 g (~0.018 mol) in 30 mL Et ₂ O	White solid; mp 48 °C	H319 eye irritation · H373 organ damage (prolonged) · H410 very toxic to aquatic life
Sulfuric acid (conc. → dil.)	Acidic work-up (protonates alkoxide)	98.08	a few mL conc. → into 7 mL water	Colorless viscous liquid; d 1.84 g/mL	H290 corrosive to metals · H314 severe skin burns & eye damage

FYI ▶ also used: anhydrous CaCl₂ (drying tube), methanol (recrystallization solvent).

GRAB FIRST (all DRY): 250-mL round-bottom · stir bar + stir plate · Claisen adapter · condenser + thermometer adapter · drying tube + cotton + anhydrous CaCl_2 · 125-mL dropping funnel · iron rings + clamps · Tygon tubing · steam bath (Part 2) · Büchner + filter flask (Part 3).

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Make PhMgBr

- 0.5 g Mg turnings + stir bar in a dry 250-mL RBF; stir vigorously 5–10 min (stopper on). Grinding strips the MgO coat.
- Pack a drying tube: cotton plug $\rightarrow$ anhydrous CaCl_2 (not tight) $\rightarrow$ cotton.
- Instructor drops in 1 small I_2 crystal (initiator).
- Build the reflux apparatus: Claisen on the RBF $\rightarrow$ straight arm gets condenser + thermometer adapter + drying tube; curved arm gets the 125-mL dropping funnel. Turn condenser water to a steady drip.
- Add solutions via the funnel off the apparatus each time (stopcock closed while filling): first 12 mL anhydrous Et_2O $\rightarrow$ run onto the Mg. Note color changes.
- Next add 2.0 mL bromobenzene + 5 mL Et_2O dropwise; the mix should self-reflux (cloudy/grey = PhMgBr forming). If it won't start: warm gently / crush Mg with a stir rod.

Part 2 — Add benzophenone

- Add 3.2 g benzophenone in 30 mL Et_2O dropwise via the funnel; reflux (steam bath) as directed.

Part 3 — Work-up + purify

- Prepare cold dilute acid: add conc. H_2SO_4 to 7 mL DI water (acid to water!). Remove drying tube; add all the cold acid to quench the alkoxide $\rightarrow \text{Ph}_3\text{C-OH}$. Conc. H_2SO_4 is extremely corrosive.
- Isolate crude solid; recrystallize from methanol (see Activity 3). Weigh $\rightarrow$ % yield; take mp / IR.

WRITE

limiting reagent = bromobenzene (0.019 mol) $\rightarrow$ theoretical $\text{Ph}_3\text{C-OH}$; % yield = actual $\div$ theoretical $\times$ 100

Experiment Notes (in lab)

Mass Mg: _____ g Vol bromobenzene: _____

Colors on adding Et_2O / bromobenzene:

Signs Grignard formed (reflux? cloudy grey?):

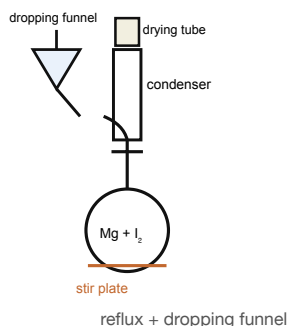
On adding benzophenone:

Work-up observations:

Crude mass / recryst. mass: _____

mp of product / % yield: _____

Setup — draw this



CONTEXT (don't copy) ▶

Why dry / drying tube? PhMgBr is a strong base — any water protonates it to benzene and it's dead. The CaCl_2 tube keeps humid air out.

Why I_2 ? a tiny bit of iodine cleans the Mg surface so the reaction starts.

Why add acid last? the first product is an alkoxide salt; H_3O^+ protonates it to the neutral alcohol (triphenylmethanol).

Clean-up & Conclusions

- Ether & organic filtrates $\rightarrow$ organic waste; acid aqueous $\rightarrow$ neutralize/aqueous waste; return glassware; bench task.
- Conclusions** (~4 sentences after lab): % yield + why (moisture killed some Grignard, transfer losses); what mp/IR confirm about purity/identity; why anhydrous conditions mattered.